

Basabraj Biswas

Aspiring Data Scientist

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PROFILE SUMMARY

Dedicated Mathematics Graduate pursuing a Master's Degree in Data Science and Artificial Intelligence. Results-driven professional with a strong foundation in mathematics and a keen interest in data science and AI, with hands-on experience in machine learning, deep learning, and business analytics.

EDUCATION

Year	Degree	Institute	CGPA/%
2024–2026	M.Sc. Data Science & AI	RKMRC, Narendrapur, University of Calcutta	7.60
2020–2024	B.Sc. Mathematics	Bangabasi College, University of Calcutta	6.904
2018–2020	Higher Secondary	Birati High School	88.8%

INTERNSHIPS

Data Science & AI Intern 15th Feb '26 – Present

Conglomerate Business Solutions Pvt. Ltd., Garia, Kolkata | Mode – On-site

- **Machine Load Trend Analysis:** Conducted EDA and anomaly detection on a 7-day live machine load dataset (40,000+ records) using an **LSTM Autoencoder** (stacked 3-layer encoder-decoder); applied Z-score and MAD threshold to identify 1,249 anomalies and delivered recommendations for predictive maintenance, auto-shutdown scheduling, and energy optimization.
- **Face Recognition & Headcount System:** Deployed a real-time face recognition and people-counting system on **Raspberry Pi 5** using **Hailo-8L NPU**; built a **GStreamer pipeline** with **SCRFD HEF** (face detection) + **ArcFace MobileFaceNet HEF** (recognition) and Hailo native tracker, persisting detection events to LanceDB with deduplication, and developed a FastAPI web dashboard for live monitoring and pipeline control.
- **Edge-Deployed Automatic Number Plate Recognition (ANPR):** Deployed a real-time **ANPR system** using **YOLOv8** on **Raspberry Pi 5** to detect and recognize vehicle number plates across multiple camera sources including **CCTV**, **IR Bullet**, and **RPi camera**.

PROJECTS

AI-Based Deepfake and Emotion Recognition System Sept – Dec '25

Team Project | Role: Model Development & Deployment

- Developed a **ResNetV2** based deepfake detection model achieving **80.77% accuracy** and emotion recognition using **ResNet50** on **FER2013** with **96.7% accuracy**, deployed via **Streamlit**.
- **DF40-HybridNet:** A **StyleGAN2**-Assisted Deepfake Detection Framework using **Xception**, **ELA**, and **LIME** for enhanced explainability.

Real-Time Anti-Money Laundering Using Temporal Graph Neural Networks (TGNNs)

- Designed an AML system using a **TGNN** to identify suspicious banking accounts by representing transactions as a **dynamic graph**, enabling network-level risk detection over traditional rule-based systems. **Real-Time Anti-Money Laundering Using Temporal Graph Neural Networks (TGNNs)** [Live Demo]

Sales Promotion Analytics and Advanced Forecasting Oct – Dec '25

Team Project at MATHNAL TECH Role: Dashboard Development & Data Visualization

- Designed interactive Power BI dashboards and utilized **SQL** and time-series models for sales forecasting and trend analysis to support business decision-making.

SKILLS

Programming & Tools: Python, R, SQL, Tableau, Latex

ML & AI: Machine Learning, Deep Learning, Time Series, Computer Vision, NLP

Soft Skills: Analytical Thinking, Problem Solving, Team Collaboration, Communication, Adaptability

CERTIFICATIONS

- Tableau A–Z: Hands-on Training for Data Science (Udemy)
- Python for Financial Analysis and Algorithmic Trading – Ongoing (Udemy)
- R Programming A–Z (Udemy)